Stress Transfer and Energy Distribution in Functionally Graded Tendon–Bone Interfaces

Federico Tremolada
MSc Biomedical Engineering
Politecnico di Milano

Abstract

The tendon–bone interface (enthesis) is a biologically optimized structure characterized by a gradual transition in mechanical properties between compliant tendon and stiff bone. This transition plays a crucial role in mitigating stress concentrations and ensuring effective load transfer.

In this study, a two-dimensional finite element model of the enthesis was developed in Abaqus/CAE to investigate the influence of different material transition laws on stress distribution and elastic energy storage. A plane stress formulation with CPS4R elements was adopted to isolate the effect of the material gradient under static uniaxial loading. Linear, exponential, and power-law gradients of the elastic modulus were compared against a sharp interface configuration.

Results show that graded material transitions significantly reduce stress concentrations and promote a more distributed load transfer. Among the tested configurations, the power-law model provides the best balance between peak stress reduction and spatial regularity of the stress field. A sensitivity analysis on the power-law exponent, evaluated through criteria of peak stress reduction and distribution regularity, identifies $n = 3$ as the optimal configuration.

Further analysis demonstrates that the distribution of strain energy density is strongly correlated with stress gradients and is primarily governed by the stiffness variation, while the effect of Poisson's ratio variation is negligible.

These findings confirm the effectiveness of functionally graded materials in improving mechanical performance at biomaterial interfaces and provide insights for the design of biomimetic implantable devices.

1 Introduction

The interface between dissimilar materials represents a critical region in many engineering and biological systems. Abrupt changes in material properties are known to generate stress concentrations, which can compromise structural integrity and lead to failure [2, 8].

In biological systems, this issue is frequently addressed through the presence of functionally graded interfaces, where material properties vary gradually across a transition region. The tendon–bone interface, known as the enthesis, is a paradigmatic example of this strategy [1, 6].

The enthesis exhibits a progressive transition in mechanical properties from the compliant tendon to the stiff bone, allowing for efficient load transfer and minimizing stress concentrations [2, 7]. This graded architecture has inspired the development of functionally graded materials (FGMs) for engineering applications, particularly in the design of biomimetic implantable devices where controlled stress transfer at dissimilar material interfaces is a primary design requirement [4].

Despite extensive experimental and theoretical studies, the mechanical role of the specific form of the material gradient remains not fully understood. In particular, the influence of different transition laws on both stress distribution and energy storage requires further investigation [3].

The aim of this study is to analyze how different material grading strategies affect the mechanical response of the tendon–bone interface. A two-dimensional finite element approach was selected as it allows systematic parametric variation of the material gradient while maintaining computational tractability and clear interpretability of results; three-dimensional and nonlinear extensions are discussed as future work. The model compares sharp and graded configurations, including linear, exponential, and power-law transitions of the elastic modulus.

In addition to stress analysis, the distribution of strain energy density is evaluated to provide a more comprehensive understanding of the mechanical behavior of the interface.

The study follows a structured multi-phase approach, including mesh convergence assessment, sensitivity analysis on the gradient exponent, and evaluation of Poisson's ratio effects, with the objective of identifying the most mechanically effective material transition law. This work contributes to the understanding of graded biological interfaces by systematically linking the form of the material gradient to stress transfer mechanisms and energy distribution.

2 Materials and Methods

2.1 Geometry and Model Setup

A two-dimensional finite element model of the tendon–bone interface (enthesis) was developed in Abaqus/CAE. The geometry consisted of a rectangular domain with a total length of 30 mm and a height of 6 mm.

The domain was divided into three regions:

- tendon (12 mm)
- enthesis (6 mm)
- bone (12 mm)

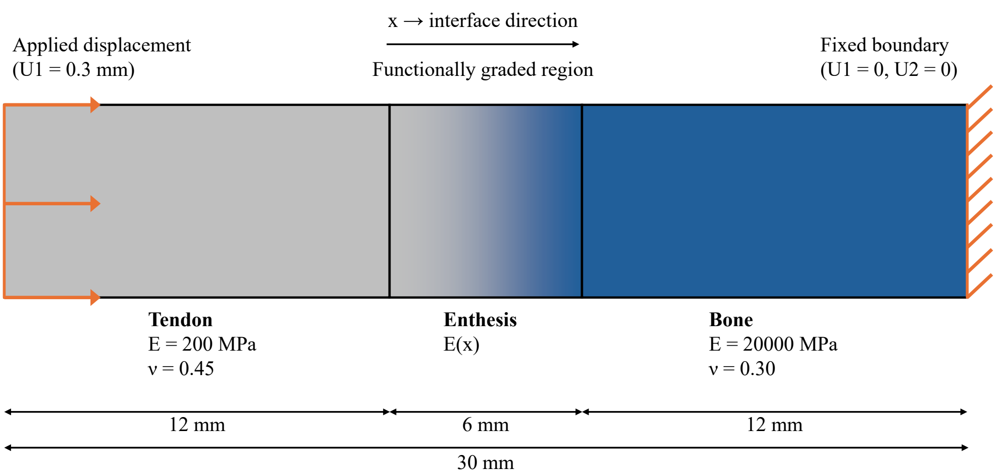

Figure 1: Schematic representation of the tendon–bone interface model. The domain is divided into tendon, enthesis, and bone regions. The enthesis is modeled as a functionally graded material, with the elastic modulus varying smoothly between tendon and bone properties according to predefined transition laws. A prescribed displacement of $U_1 = 0.3$ mm is applied on the tendon side, while the bone side is fully constrained ($U_1 = U_2 = 0$).

The model was implemented as a single 2D deformable part with longitudinal partitions. The enthesis region was discretized into multiple layers to approximate a continuous variation of material properties.

2.2 Material Models

All materials were modeled as linear elastic and isotropic.

The tendon was assigned a Young's modulus of 200 MPa and a Poisson's ratio of 0.45, representing its nearly incompressible behavior.

The bone was assigned a Young's modulus of 20,000 MPa and a Poisson's ratio of 0.30.

The resulting stiffness ratio between bone and tendon is approximately 100:1, introducing a strong mechanical mismatch at the interface.

2.2.1 Elastic modulus transition laws

To describe the variation of material properties across the enthesis, a local longitudinal coordinate x was defined within the transition region, with $x = 0$ at the tendon side and $x = L_e$ at the bone side, where $L_e = 6$ mm is the enthesis length.

The tendon and bone elastic moduli were defined as:

$$E_t = 200 \text{ MPa}, \qquad E_b = 20{,}000 \text{ MPa}$$

Different transition laws were used to assign the elastic modulus $E(x)$ within the enthesis region.

Sharp interface In the sharp interface model, no transition region was introduced. The elastic modulus was defined as a discontinuous function:

$$E(x) = \begin{cases} E_t, & x < x_i \\ E_b, & x \geq x_i \end{cases}$$

where $x_i = 0$ corresponds to the tendon–bone interface location (i.e., the discontinuity is placed at the tendon-side boundary of the enthesis region, at $x = 12$ mm in global coordinates).

Linear gradient For the linear graded model, the elastic modulus varied linearly across the enthesis:

$$E(x) = E_t + (E_b - E_t)\frac{x}{L_e}$$

Exponential gradient For the exponential graded model, the elastic modulus was defined as:

$$E(x) = E_t \left(\frac{E_b}{E_t}\right)^{x/L_e}$$

This formulation ensures that $E(0) = E_t$ and $E(L_e) = E_b$.

Power-law gradient For the power-law model, the elastic modulus was defined as:

$$E(x) = E_t + (E_b - E_t)\left(\frac{x}{L_e}\right)^n$$

where n is the exponent controlling the shape of the stiffness transition. The exponent n governs the spatial distribution of stiffness across the enthesis: lower values result in a more gradual increase from tendon to bone, while higher values shift the stiffening towards the bone side, leading to a delayed but sharper transition. The sensitivity analysis on the exponent is described in Section 2.4.

Layer-wise discretization Since the model was implemented in Abaqus/CAE using a finite number of partitions, the continuous functions $E(x)$ were approximated through layer-wise discretization. Each layer was assigned the elastic modulus evaluated at its centroid position within the enthesis region.

2.3 Finite Element Implementation

A plane stress formulation was adopted, consistent with the thin rectangular geometry of the domain and the applied uniaxial loading condition, which minimizes out-of-plane stress components. CPS4R elements (4-node quadrilateral elements with reduced integration) were used.

A static general step was employed, with geometric nonlinearity disabled (NLGEOM = OFF), in order to isolate the effect of material property variation on the stress field.

Boundary conditions were applied as follows:

- the right boundary (bone side) was fully constrained ($U_1 = 0$, $U_2 = 0$)
- a prescribed displacement of $U_1 = 0.3$ mm was applied on the left boundary (tendon side), with U_2 left free

This configuration generates a uniaxial tensile state along the longitudinal direction.

A structured mesh was adopted, with local refinement in the enthesis region (element size $\approx$ 0.2–0.25 mm) and a coarser discretization in the tendon and bone regions, in order to accurately capture stress gradients while maintaining computational efficiency.

2.4 Parametric Study and Model Selection

A structured multi-phase parametric analysis was conducted to identify the most mechanically effective material configuration, following a progressive refinement strategy.

Phase A — Discretization refinement

The number of layers in the enthesis region was varied (8, 12, and 16 layers) in order to assess the convergence of the discretized material gradient. Convergence was evaluated by monitoring the peak S11 value and the smoothness of the stress profile along the central axis. The 16-layer configuration was selected as it provided a stable, mesh-independent stress distribution and was therefore adopted for all subsequent analyses.

Phase B — Power-law exponent sensitivity

A sensitivity analysis was performed on the exponent of the power-law distribution ($n = 1.5, 2, 3, 5$), with 16 layers in the enthesis region. The objective was to evaluate how the shape of the stiffness gradient influences stress transfer across the interface.

The selection criteria were:

- reduction of peak stress magnitude
- spatial regularity and symmetry of the stress distribution
- physical consistency of load transfer across the interface

Based on these criteria, the configuration with $n = 3$ and 16 layers was identified as the optimal compromise between stress reduction and distribution smoothness.

Phase C — Influence of Poisson's ratio

Two variants of the optimal configuration were investigated to assess the influence of transverse deformation on the mechanical response:

- M17: power-law ($n = 3$, 16 layers) with constant Poisson's ratio throughout the enthesis
- M18: power-law ($n = 3$, 16 layers) with spatially varying Poisson's ratio

2.4.1 Poisson's ratio distribution

In M17, Poisson's ratio was held constant at $\nu = 0.35$ throughout the entire enthesis region.

In M18, a gradual spatial variation was introduced to ensure continuity between tendon and bone properties, with boundary values:

$$\nu_t = 0.45, \qquad \nu_b = 0.30$$

A power-law distribution consistent with the elastic modulus formulation was adopted:

$$\nu(x) = \nu_t + (\nu_b - \nu_t)\left(\frac{x}{L_e}\right)^n$$

with $n = 3$,implemented using the same layer-wise discretization adopted for the elastic modulus. This formulation ensures mechanical consistency between normal and transverse deformation fields across the graded region.

Phase D — Energy-based validation

An additional validation phase was conducted through the analysis of strain energy density (SENER), in order to evaluate the global mechanical response and the spatial distribution of stored elastic energy.

2.5 Summary of Investigated Configurations

Table 1: Summary of the main model configurations considered in the study.

Model	Transition type	Layers	Notes
Sharp interface	Discontinuous	–	Baseline reference
Linear gradient	Linear	8	Progressive stiffness increase
Exponential gradient	Exponential	8	Rapid stiffening near bone
Power-law	Power-law ($n = 1.5$–5)	16	Sensitivity analysis (Phase B)
M17	Power-law ($n = 3$)	16	Constant Poisson's ratio
M18	Power-law ($n = 3$)	16	Variable Poisson's ratio

2.6 Data Extraction and Post-Processing

Stress and energy data were extracted along the central line of the model (mid-height of the domain, $y \approx 3$ mm) using custom Python scripts interfacing with Abaqus output databases (.odb). The scripts computed element-wise values of S11 and SENER, identified global extrema, and exported profiles in CSV format for comparative analysis.

The evaluation focused on: longitudinal stress (S11) distribution and peak values, spatial gradient and curvature of S11, and strain energy density (SENER) distribution and its spatial correlation with the stress field.

3 Results

3.1 Stress Distribution (S11) — Phase 1

The longitudinal stress distribution (S11) along the central axis of the model is shown in Figure 2 for the different material transition laws considered in Phase 1 (sharp, linear, exponential, power-law with $n = 0.5$ and $n = 2$).

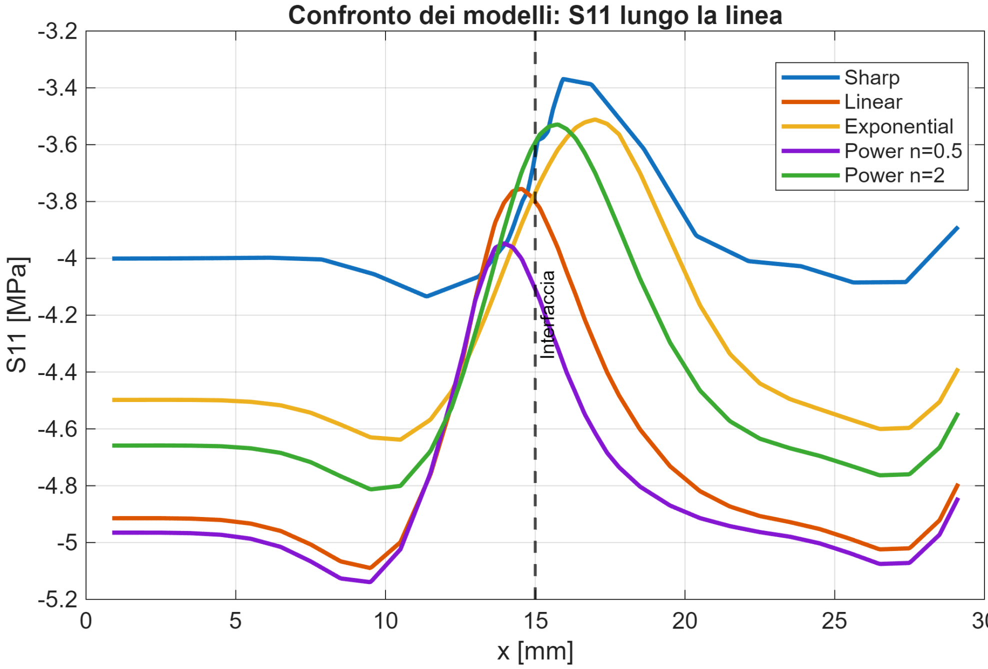

Figure 2: Longitudinal stress distribution (S11) along the central axis for different material transition laws (Phase 1). The graded configurations reduce stress concentration and redistribute the load transfer across the enthesis region compared to the sharp interface. The dashed vertical line indicates the nominal tendon–bone interface at $x = 12$ mm.

The sharp interface configuration exhibits a pronounced stress peak localized at the nominal tendon–bone interface, reflecting the abrupt stiffness mismatch between the two regions.

All graded configurations reduce the peak stress magnitude. The linear and exponential models achieve moderate smoothing but introduce asymmetries in the stress profile: the linear gradient shifts the peak towards the tendon side, while the exponential gradient produces a response biased towards the bone side. The power-law model with $n = 2$ provides the most balanced response among Phase 1 configurations, maintaining a smooth and nearly symmetric distribution. The role of the exponent is further investigated in the sensitivity analysis (Section 3.5).

3.2 Stress Gradient Analysis

The spatial derivative of S11 along the central axis is shown in Figure 3.

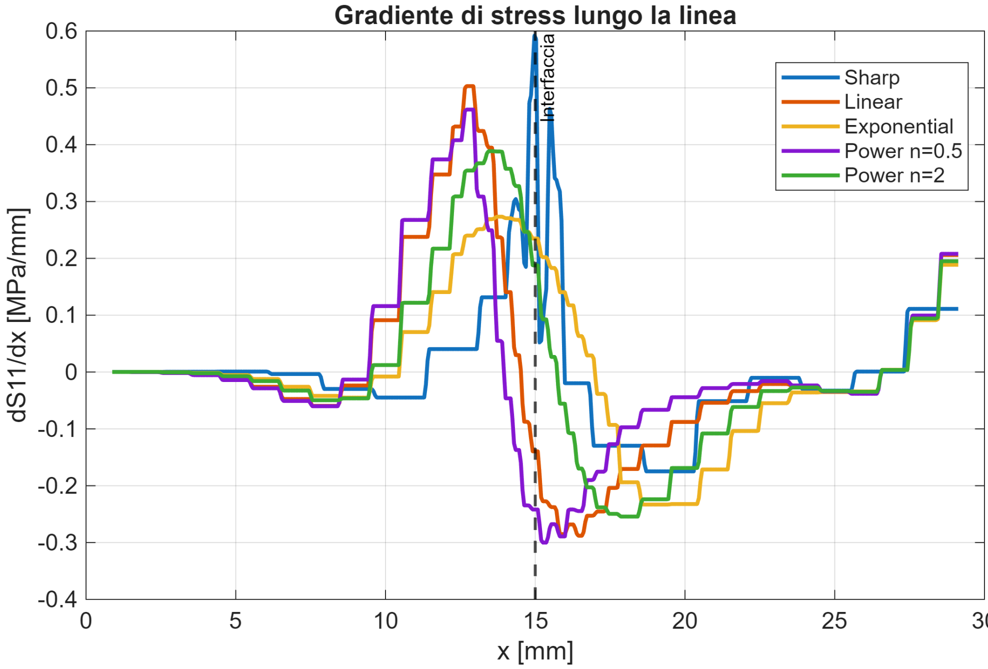

Figure 3: Spatial gradient of S11 ($\mathrm{d}S_{11}/\mathrm{d}x$) along the central axis for the different material models. Graded configurations reduce the peak gradient intensity and distribute the load transfer over a wider region compared to the sharp interface. Power-law models yield a more symmetric redistribution than linear or exponential transitions.

The sharp interface model exhibits a highly concentrated gradient peak at the interface, confirming an abrupt load transfer. All graded configurations distribute the gradient over a wider spatial region. The power-law models provide the most symmetric redistribution, while the linear and exponential models show more asymmetric profiles. This asymmetry motivates the adoption of the power-law form as the basis for the subsequent optimization.

3.3 Stress Curvature Analysis

The second spatial derivative of S11, which quantifies the local curvature of the stress field, is shown in Figure 4.

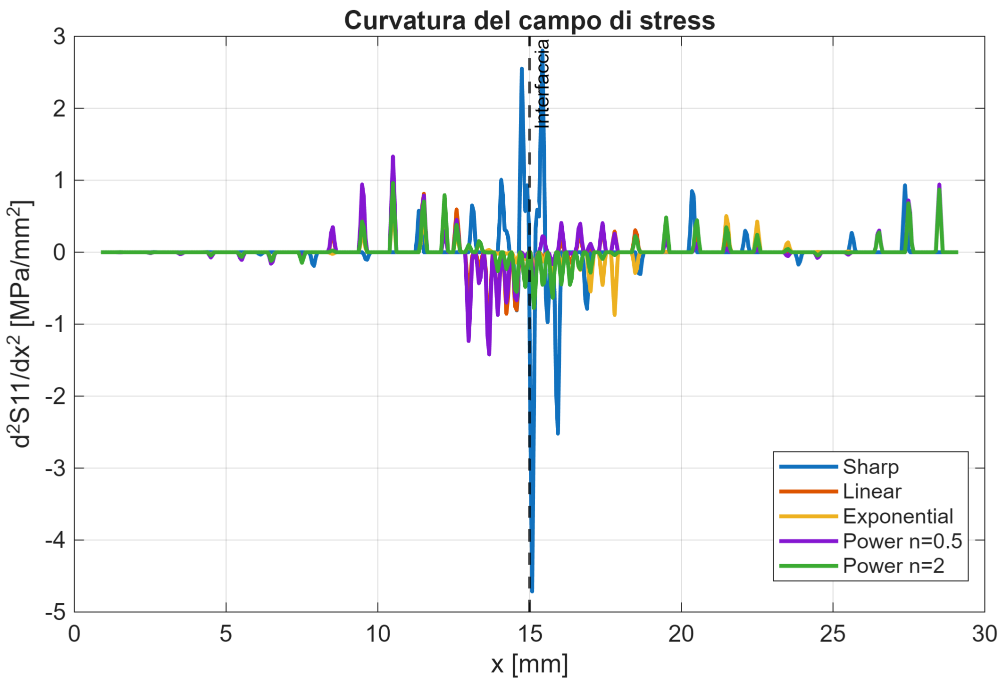

Figure 4: Second spatial derivative of S11 (stress curvature) along the central axis. Graded models reduce curvature peaks and broaden their spatial extent, indicating a smoother and more stable stress field. The power-law model with $n = 2$ yields the most regular curvature profile in Phase 1.

The sharp interface exhibits strong, localized curvature peaks at the interface. All graded models reduce both the magnitude and spatial localization of these peaks. The power-law model with $n = 2$ provides the most regular curvature distribution in Phase 1, with the lowest oscillation amplitude. Lower exponents (e.g., $n = 0.5$) introduce local oscillations that suggest a less stable stress field. These observations, combined with the gradient analysis, informed the selection of the power-law form for the sensitivity study.

3.4 Quantitative Comparison — Phase 1

Table 2: Qualitative summary of S11 metrics for the material transition laws considered in Phase 1. Numerical peak values are not reported as they depend on the specific layer discretization used in this phase (8 layers); quantitative comparison is performed in Phase 2 with the refined 16-layer model.

Model	Peak S11	Peak position	Stress profile
Sharp interface	High	At interface	Highly localized
Linear gradient	Reduced	Tendon side	Moderate smoothing
Exponential gradient	Reduced	Bone side	Asymmetric
Power-law ($n = 0.5$)	Strongly reduced	Shifted	Irregular
Power-law ($n = 2$)	Moderately reduced	Stable	Balanced

The results highlight that peak reduction alone is not a sufficient criterion for model selection. The spatial regularity and symmetry of the stress profile must also be considered.

3.5 Power-Law Exponent Sensitivity Analysis

A sensitivity analysis was performed on the power-law exponent ($n = 1.5, 2, 3, 5$) using 16 layers in the enthesis region, with all other parameters held constant.

Increasing n progressively shifts the stiffness transition towards the bone side, redistributing the stress field across the enthesis. Lower values of n retain relatively high stress concentrations near the interface, while higher values promote a broader load transfer. However, for $n = 5$, the stress distribution becomes less regular, with a shift of the peak towards the bone region and the emergence of localized oscillations, suggesting that an excessively steep gradient concentrates load near the stiff region.

Among the configurations analyzed, $n = 3$ provides the best compromise: it achieves a significant reduction of the stress peak while preserving a spatially regular and symmetric distribution. The power-law model with $n = 3$ and 16 layers was therefore selected for all subsequent analyses.

3.6 Refined Analysis: M17 vs. M18

The two refined configurations — M17 (constant Poisson's ratio) and M18 (spatially varying Poisson's ratio) — were compared to isolate the influence of transverse deformation on the mechanical response.

3.6.1 Stress Distribution

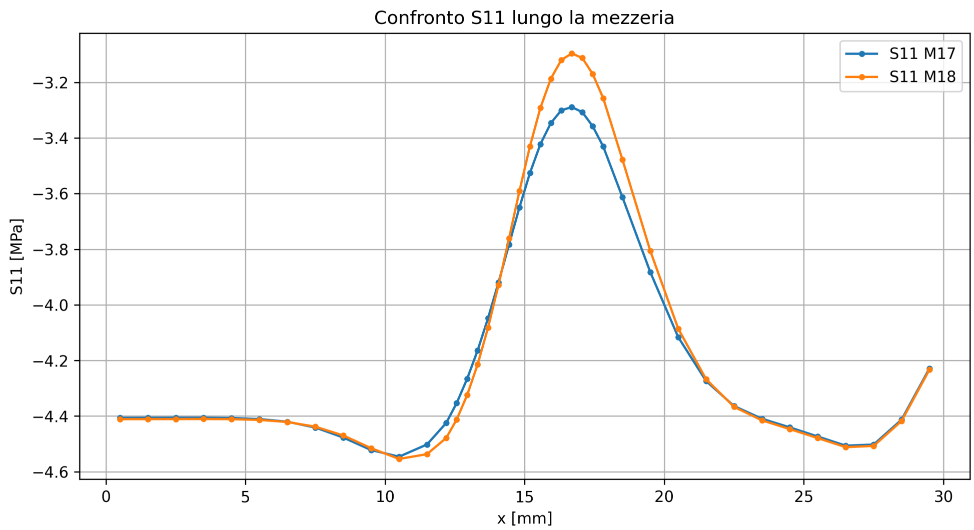

Figure 5: Longitudinal stress distribution (S11) along the central axis for M17 and M18 (power-law, $n = 3$, 16 layers). The two profiles are nearly coincident, with a stress peak of approximately -3.25 MPa at $x \approx 15$ mm, confirming the negligible influence of Poisson's ratio variation on the longitudinal stress field.

The two models exhibit nearly identical stress distributions. Only minor differences are observed in peak magnitude and local sharpness, confirming that the longitudinal stress field is largely governed by the elastic modulus gradient and is insensitive to transverse deformation effects within the linear elastic framework adopted here.

3.6.2 Elastic Energy Distribution (SENER)

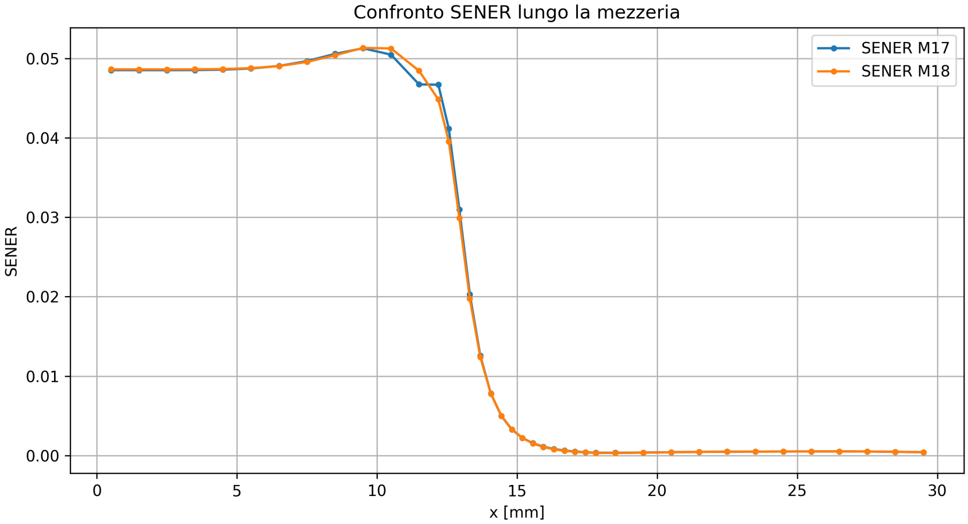

Figure 6: Strain energy density (SENER) distribution along the central axis for M17 and M18. Energy is concentrated within the enthesis region (approximately $0 \leq x \leq 15$ mm), reaching a peak of approximately 0.053 J/mm^3 near the tendon-side boundary of the enthesis. The profiles for constant and variable Poisson's ratio are nearly overlapping.

Both models show a concentration of elastic energy within the enthesis region, with a rapid drop at the transition to bone. The SENER profiles are virtually coincident, with differences in peak values below 0.1%.

3.6.3 Stress–Energy Relationship

A strong spatial correlation is observed between the stress gradient and the energy density field: the maximum SENER accumulation occurs in the region of highest stress variation (near the tendon-side boundary of the enthesis, $x \approx 10$–15 mm), rather than at the location of the absolute stress peak. This indicates that elastic energy storage is driven by the combined stress–strain field and is particularly sensitive to local stiffness gradients. The graded configuration thus promotes a distributed energy absorption mechanism, which is mechanically advantageous compared to the localized energy concentration characteristic of the sharp interface.

3.6.4 Quantitative Energy Metrics

Total elastic energy and peak SENER values are virtually identical for M17 and M18 (differences $< 0.1\%$). A marginal smoothing of the energy gradient is observed in M18, without meaningful impact on the overall mechanical response. These results confirm that the elastic modulus gradient is the dominant parameter controlling both stress distribution and energy storage.

4 Discussion

4.1 Role of the Material Transition Law

The comparison between different grading laws confirms that the specific functional form of the material transition is a primary determinant of the mechanical behavior of the interface. A sharp interface concentrates stress over a very short spatial region as a direct consequence of the abrupt 100:1 stiffness mismatch, generating a localized peak in both the stress field and its spatial derivatives.

Functionally graded configurations mitigate this effect by distributing the load transfer across the enthesis region. However, not all gradient forms perform equivalently: the linear and exponential models introduce spatial asymmetry in the stress profile, shifting the peak towards the tendon or bone side respectively. The power-law distribution offers a more controllable modulation of the stiffness gradient, as the exponent n provides an explicit parameter to tune the balance between early and late stiffening within the transition region.

The sensitivity analysis reveals a non-monotonic relationship between n and mechanical performance: while increasing n reduces peak stress and broadens load transfer, excessively high exponents (e.g., $n = 5$) shift the stiffness transition towards the bone side and introduce localized irregularities. The optimal value $n = 3$ represents the exponent at which the gradient is steep enough to achieve significant peak reduction while remaining smooth enough to avoid stress re-concentration near the bone region.

4.2 Stress–Energy Relationship and Mechanical Implications

The distribution of strain energy density provides complementary information beyond peak stress values. The observation that maximum SENER accumulation occurs in the region of highest stress gradient — rather than at the absolute stress peak — highlights the importance of stiffness continuity in governing energy storage. In the sharp interface, this energy is concentrated in a very small volume, increasing the risk of damage initiation. In the graded configuration, energy is distributed over the full enthesis length, reducing local energy density and increasing structural robustness.

This mechanism is consistent with the proposed role of the enthesis as a functional buffer between dissimilar tissues [2, 7], and supports the use of energy-based metrics alongside stress-based criteria in the evaluation of graded interface performance.

4.3 Influence of Poisson's Ratio

The comparison between M17 and M18 shows that a spatially varying Poisson's ratio has a negligible influence on both longitudinal stress and energy distributions (differences $< 0.1\%$). Within the linear elastic, plane stress framework adopted here, transverse deformation effects are secondary to the stiffness gradient under uniaxial loading. This finding simplifies the design space for biomimetic FGM interfaces: accurate specification of the elastic modulus profile is the critical design parameter, while Poisson's ratio variation — which is more difficult to control in manufactured materials — has minimal mechanical consequence in this loading regime.

4.4 Consistency with FGM Theory and Literature

These results are consistent with the established theory of functionally graded materials, which predicts stress concentration reduction through smooth material transitions [8]. The superiority of non-linear (power-law) over linear gradients in controlling stress distribution has been documented in both computational and experimental studies on graded biomaterials [4, 3], and the present results provide further quantitative support for this finding in the specific context of the tendon–bone interface.

4.5 Limitations and Future Work

The model involves several intentional simplifications. The 2D plane stress formulation neglects out-of-plane stress components and three-dimensional geometric effects present in the actual enthesis. All materials are assumed linear elastic and isotropic, whereas biological tissues are known to exhibit anisotropy, viscoelasticity, and nonlinear behavior. The layer-wise discretization introduces a stepped approximation of the continuous gradient. Finally, only static loading was considered.

These simplifications were adopted to isolate the effect of the material gradient in a controlled parametric framework. Future work should extend the model to three-dimensional geometries, incorporate tissue-specific constitutive laws (e.g., fiber-reinforced hyperelastic models for tendon), and include cyclic loading conditions to evaluate fatigue behavior.

5 Conclusions

This study systematically investigated the mechanical behavior of the tendon–bone interface through a parametric finite element analysis of functionally graded material transitions. The main findings are:

- Graded transitions significantly reduce stress concentrations compared to a sharp interface, with the power-law distribution outperforming linear and exponential alternatives in terms of both peak reduction and spatial regularity.
- Among the power-law configurations, $n = 3$ provides the optimal balance between stress peak reduction and profile regularity; higher exponents produce stress re-concentration near the bone region.
- Strain energy density is governed by the stiffness gradient rather than the absolute stress magnitude, and the graded configuration promotes distributed energy absorption over the enthesis region.
- Poisson's ratio spatial variation has negligible influence on the mechanical response under the adopted loading conditions, identifying the elastic modulus gradient as the dominant design parameter.

These findings provide quantitative guidelines for the design of biomimetic interfaces and functionally graded structures in biomedical applications where controlled stress transfer between dissimilar materials is required.

References

[1] S. Thomopoulos, G. R. Williams, J. A. Gimbel, M. Favata, and L. J. Soslowsky, "Variation of biomechanical, structural, and compositional properties along the tendon to bone insertion site," *Journal of Orthopaedic Research*, vol. 21, no. 3, pp. 413–419, 2003.

[2] G. M. Genin, A. Kent, V. Birman, B. Wopenka, J. D. Pasteris, P. J. Marquez, and S. Thomopoulos, "Functional grading of mineral and collagen in the attachment of tendon to bone," *Biophysical Journal*, vol. 97, no. 4, pp. 976–985, 2009.

[3] S. Kuznetsov, M. Pankow, K. Peters, and H.-Y. S. Huang, "A structural-based computational model of tendon–bone insertion tissues," *Mathematical Biosciences*, vol. 327, 108411, 2020.

[4] J. M. Lowen and J. K. Leach, "Functionally graded biomaterials for use as model systems and replacement tissues," *Advanced Functional Materials*, vol. 30, 1909089, 2020.

[5] C. Wang, X. Zhang, D. M. Wang, P. S. H. Yung, R. S. Tuan, and D. F. E. Ker, "Optimized design of an enthesis-mimicking suture anchor-tendon hybrid graft for mechanically robust bone-tendon repair," *Acta Biomaterialia*, vol. 176, pp. 277–292, 2024.

[6] M. Benjamin, T. Kumai, S. Milz, B. M. Boszczyk, A. A. Boszczyk, and J. R. Ralphs, "The skeletal attachment of tendons – tendon 'entheses'," *Comparative Biochemistry and Physiology Part A*, vol. 133, no. 4, pp. 931–945, 2002.

[7] J. Apostolakos, T. J. S. Durant, C. R. Dwyer, R. P. Russell, J. H. Weinreb, F. Alaee, K. Beitzel, M. B. McCarthy, M. P. Cote, and A. D. Mazzocca, "The enthesis: a review of the tendon-to-bone insertion," *Muscles, Ligaments and Tendons Journal*, vol. 4, no. 3, pp. 333–342, 2014.

[8] S. Suresh, "Graded materials for resistance to contact deformation and damage," *Science*, vol. 292, pp. 2447–2451, 2001.

[9] M. Koizumi, "FGM activities in Japan," *Composites Part B: Engineering*, vol. 28, pp. 1–4, 1997.